Phase 1 - Synopsis

Heart Disease Prediction Using ML

Problem Id: 13

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- **Problem Description:**

Heart disease is one of the leading causes of death globally, and early detection plays a critical role in reducing mortality. Machine Learning (ML) techniques can assist in predicting the likelihood of heart disease based on patient clinical data.

This project focuses on designing an end-to-end ML classification system to predict heart disease risk using patient health parameters. The system processes raw medical data, applies preprocessing techniques, selects relevant features, and trains multiple classification models to evaluate predictive performance.

- **Steps of Implementation:**

1. Data Collection

A heart disease dataset containing patient clinical features was used. The target variable indicates whether heart disease is present (1) or not (0).

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2. Data Exploration

The dataset was analyzed using summary statistics and visualizations (histograms, heatmaps) to understand feature distributions and relationships.

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3. Data Cleaning

Missing values were identified and handled using:

- Mean/median for numerical features
- Mode for categorical features

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4. Data Preprocessing

- **Encoding:** Converted categorical variables into numerical form
- **Scaling:** Applied StandardScaler to normalize feature values

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5. Feature Selection

Relevant features were selected using correlation analysis to improve model performance and reduce redundancy.

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6. Train-Test Split

Dataset was split into:

- 80% training data
- 20% testing data

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7. Model Training

Three models were trained:

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)

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8. Model Evaluation

Models were evaluated using:

- Accuracy, Precision, Recall, F1-score
- Confusion Matrix
- ROC Curve

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9. Result Analysis

Model performances were compared and the best-performing model was identified.

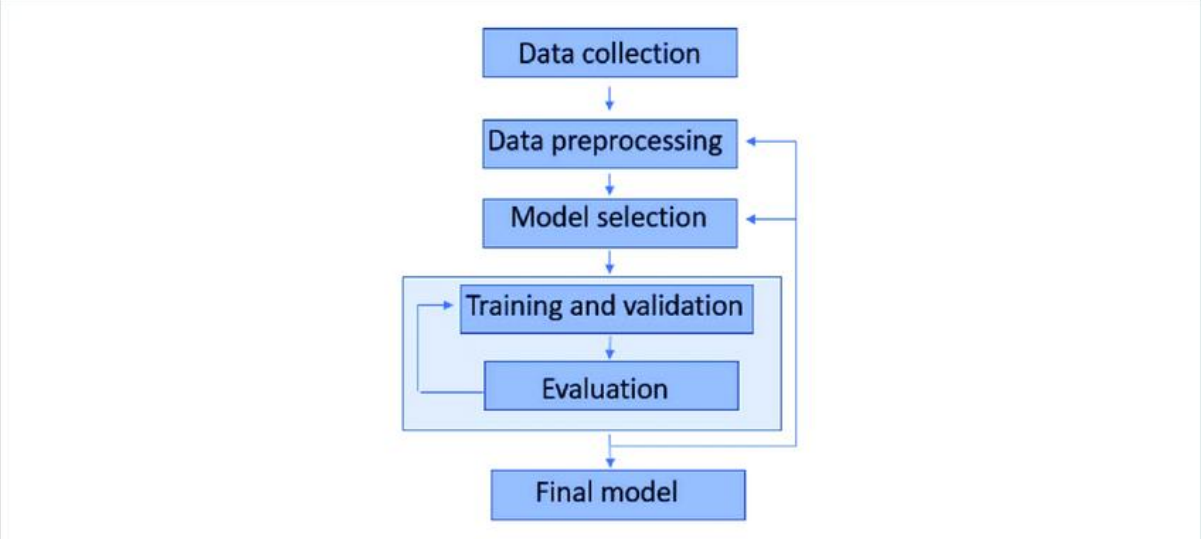

Fig. 1 architecture

- **Expected Modules/ Libraries to be used :**

1. NumPy

Used for numerical computations and handling arrays. It provides efficient operations for mathematical calculations on datasets.

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2. Pandas

Used for data manipulation and analysis. It helps in loading datasets, handling missing values, and organizing data using DataFrames.

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3. Matplotlib

Used for data visualization such as plotting graphs, histograms, and ROC curves to understand data distribution and model performance.

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4. Seaborn

Built on Matplotlib, used for advanced visualizations like heatmaps and correlation matrices for better data interpretation.

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5. Scikit-learn

Main library used for implementing machine learning models and preprocessing techniques.

Key modules from Scikit-learn:

- **model_selection** → Splitting dataset into training and testing sets
- **preprocessing** → Scaling and encoding features
- **linear_model** → Logistic Regression implementation
- **neighbors** → KNN classifier
- **svm** → Support Vector Machine
- **metrics** → Evaluation metrics like accuracy, precision, recall, F1-score

- **Expected Evaluation metrics and Plots to be used:**

1. Evaluation Metrics

1.1 Accuracy

Measures the overall correctness of the model.
It gives the ratio of correctly predicted instances to total instances.

Why used:
Provides a general idea of model performance, but may be misleading if data is imbalanced.

1.2 Precision

Measures how many predicted positive cases are actually correct.

Why used:
Important to reduce **false positives**, especially when predicting disease unnecessarily.

1.3 Recall (Sensitivity)

Measures how many actual positive cases are correctly identified.

Why used:
Critical in healthcare because missing a heart disease case (false negative) is far worse than a false alarm.

1.4 F1-Score

Harmonic mean of precision and recall.

Why used:
Balances precision and recall, especially useful when both false positives and false negatives matter.

1.5 Confusion Matrix

Displays counts of:

- True Positives (TP)
- True Negatives (TN)
- False Positives (FP)
- False Negatives (FN)

Why used:
Gives a complete breakdown of model predictions instead of a single number.

1.6 ROC Curve and AUC Score

- ROC curve plots True Positive Rate vs False Positive Rate
- AUC represents model's ability to distinguish between classes

Why used:
Helps evaluate model performance across different thresholds, not just one fixed cutoff.

2. Visualization Plots

2.1 Histogram / Distribution Plots

Displays distribution of individual features.

Why used:
Helps detect skewness, outliers, and data spread.

2.2 Confusion Matrix Heatmap

Visual representation of prediction results.

Why used:
Makes it easier to interpret classification errors.

2.3 ROC Curve

Shows trade-off between sensitivity and specificity.

Why used:
Helps compare multiple models and select the best one.

- **Required topics from the subject:**

1. Data Preprocessing
Data preprocessing involves cleaning and transforming raw data into a suitable format for machine learning models. It includes handling missing values, encoding categorical variables, and scaling features. This step ensures better model accuracy and reliability.

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2. Feature Selection
Feature selection is the process of identifying the most relevant input variables for the model. It helps reduce dimensionality, improve performance, and avoid overfitting. Techniques like correlation analysis are commonly used.

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3. Supervised Learning
Supervised learning is a type of machine learning where the model is trained using labeled data. In this project, classification algorithms like Logistic Regression, KNN, and SVM are used to predict heart disease.

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4. Classification Algorithms
Classification algorithms are used to categorize data into predefined classes. This project uses multiple classifiers to predict whether a patient has heart disease (1) or not (0).

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5. Model Evaluation Techniques
Model evaluation involves assessing the performance of trained models using metrics like accuracy, precision, recall, and F1-score. It ensures that the model performs well on unseen data.

6. Data Visualization

Data visualization uses graphs and plots to understand data patterns and model results. Techniques like heatmaps, histograms, and ROC curves help in better interpretation.

7. Train-Test Splitting

The dataset is divided into training and testing sets to evaluate model performance. This prevents overfitting and ensures the model generalizes well to new data.

8. Overfitting and Underfitting

Overfitting occurs when a model learns noise from training data, while underfitting happens when it fails to learn patterns. Proper model selection and validation help balance both.

- **Platform Required:**

Visual Code studio, jupyter notebook, Kaggle

- **Books and Link Sources:**

For Dataset : https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction

For understanding the libraries and syntax : https://www.geeksforgeeks.org/

For reference projects: https://github.com/

Book : **Introduction to Machine Learning with Python**